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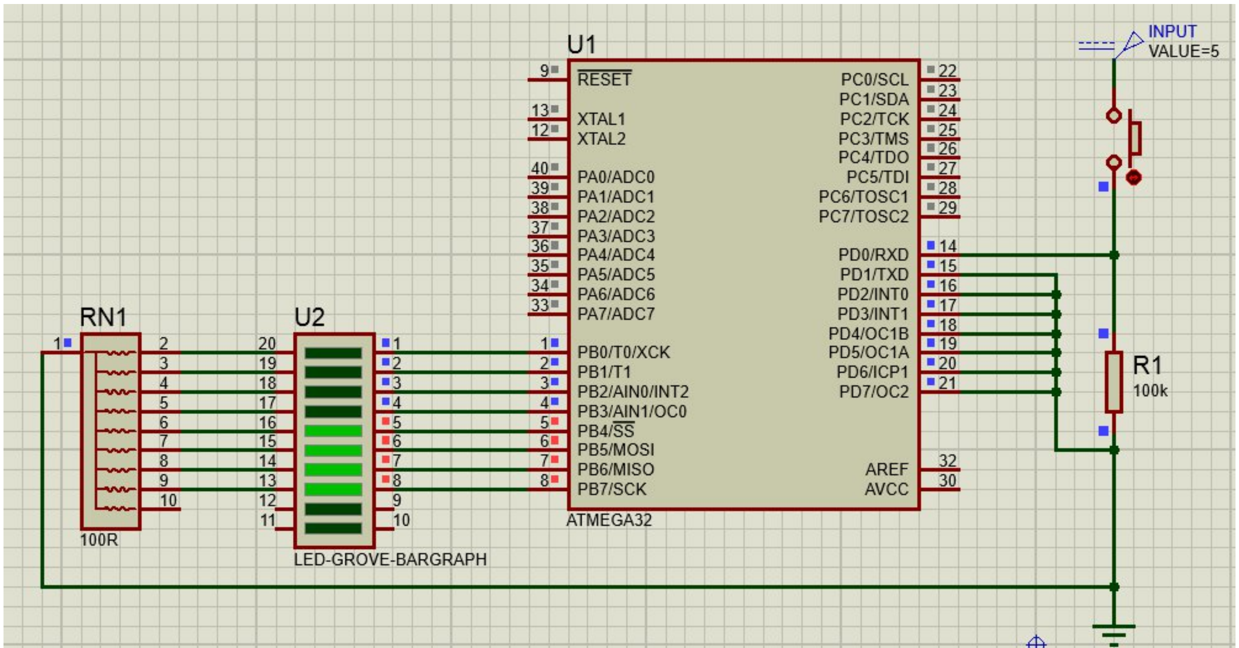
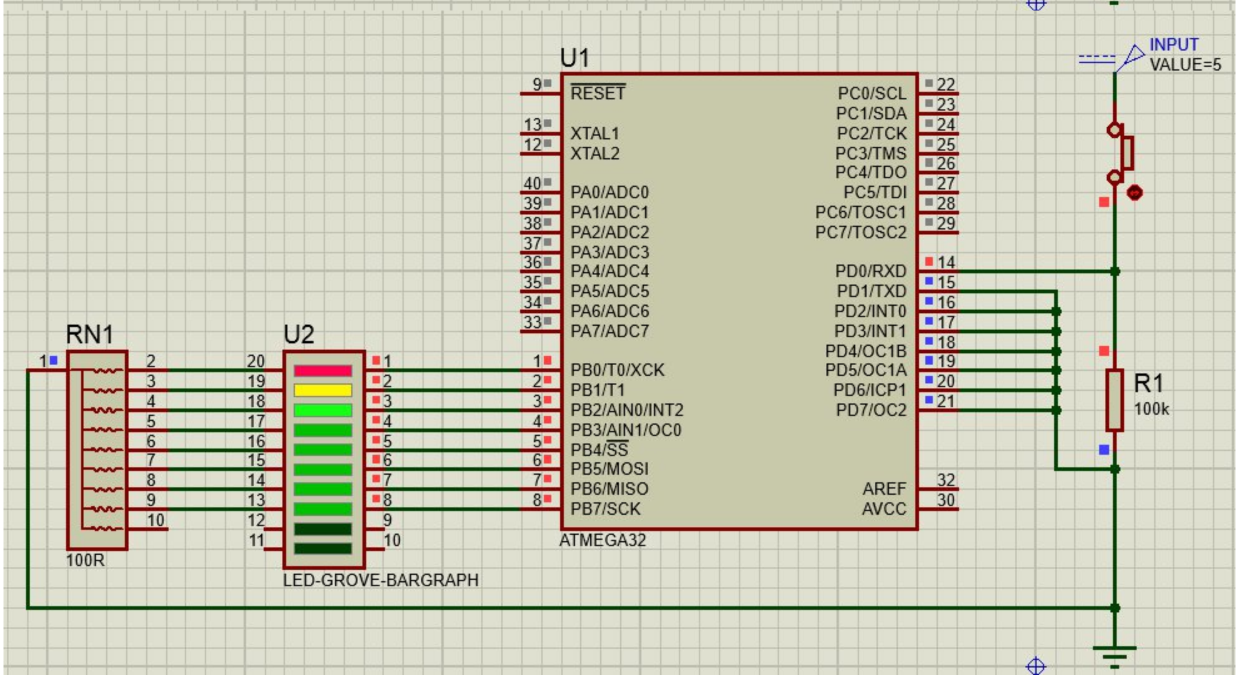
Day 08 – 19.09.2025

ME 2261 - Embedded Systems Log Book

Task 1

```
#include <avr/io.h>

int main(void)
{
    DDRD = 0x00;    // set PORTD for Input
    DDRB = 0xFF;    // set PORTB for Output
    PORTB = 0x00;   // turn off all LEDs initially
    while(1){
    if (PIND == 1)  // PIND == 0x01, PD0 is checked
    {
        PORTB = 0xFF;
    }
    else
    {
        PORTB = 0xF0;
    }
    }
    return 1;
}
```

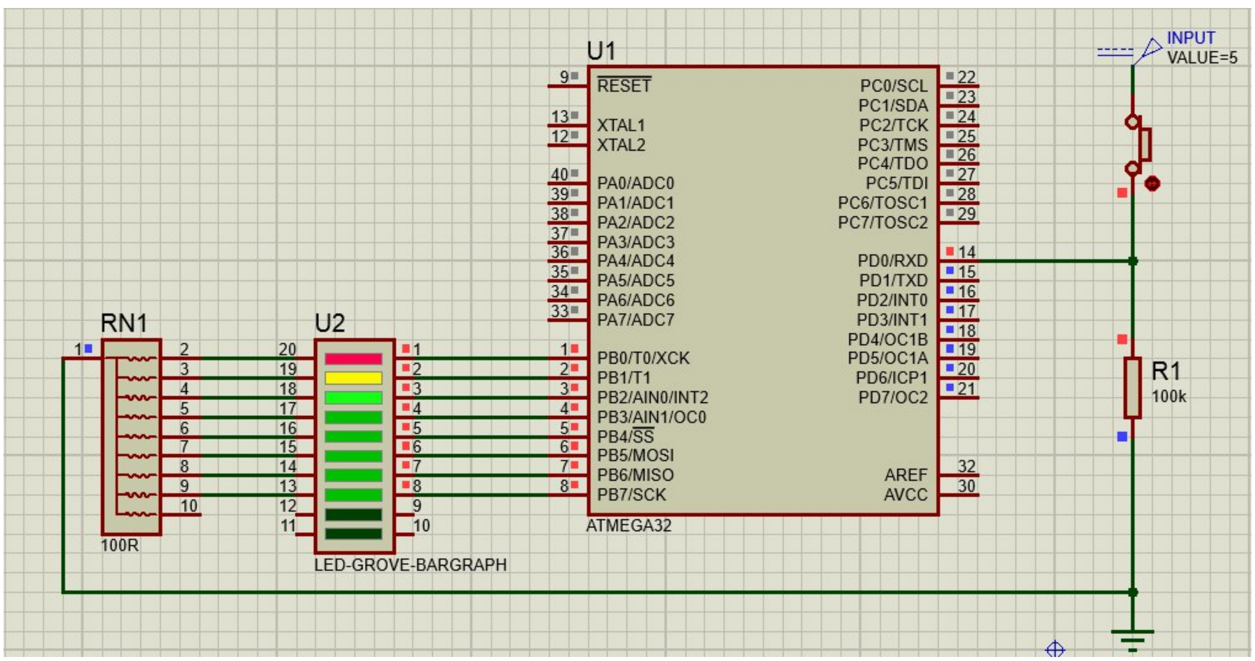

U1
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
ATMEGA32
RN1
100R
U2
LED-GROVE-BARGRAPH
R1
100k
INPUT
VALUE=5


U1
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
ATMEGA32
RN1
100R
U2
LED-GROVE-BARGRAPH
R1
100k
INPUT
VALUE=5

Task 2

To avoid connecting every pins with ground.

```c
#include <avr/io.h>

int main(void)
{
    DDRD = 0b11111110;  // set PORTD.0 pin only Input
    DDRB = 0xFF;    // set PORTB for Output
    PORTB = 0x00;   // turn off all LEDs initially
    while(1){
    if (PIND == 1)  // PIND == 0x01, PD0 is checked
    {
        PORTB = 0xFF;
    }
    else
    {
        PORTB = 0xF0;
    }
    }
    return 1;
}
```



Task 3

Switch-case statement

```c
#include <avr/io.h>

int main(void)
{
    DDRB = 0xFF;    // set PORTB for Output
    unsigned char digit = '2';  // input
    int led_pattern = 0x00; // output

    switch (digit){
        case '1': led_pattern = 0b00001111; break;
        case '2': led_pattern = 0b11000011; break;
        case '3': led_pattern = 0b01010101; break;
        default: led_pattern = 0b11111111;
    }
    PORTB = led_pattern;
    return 1;
}
```

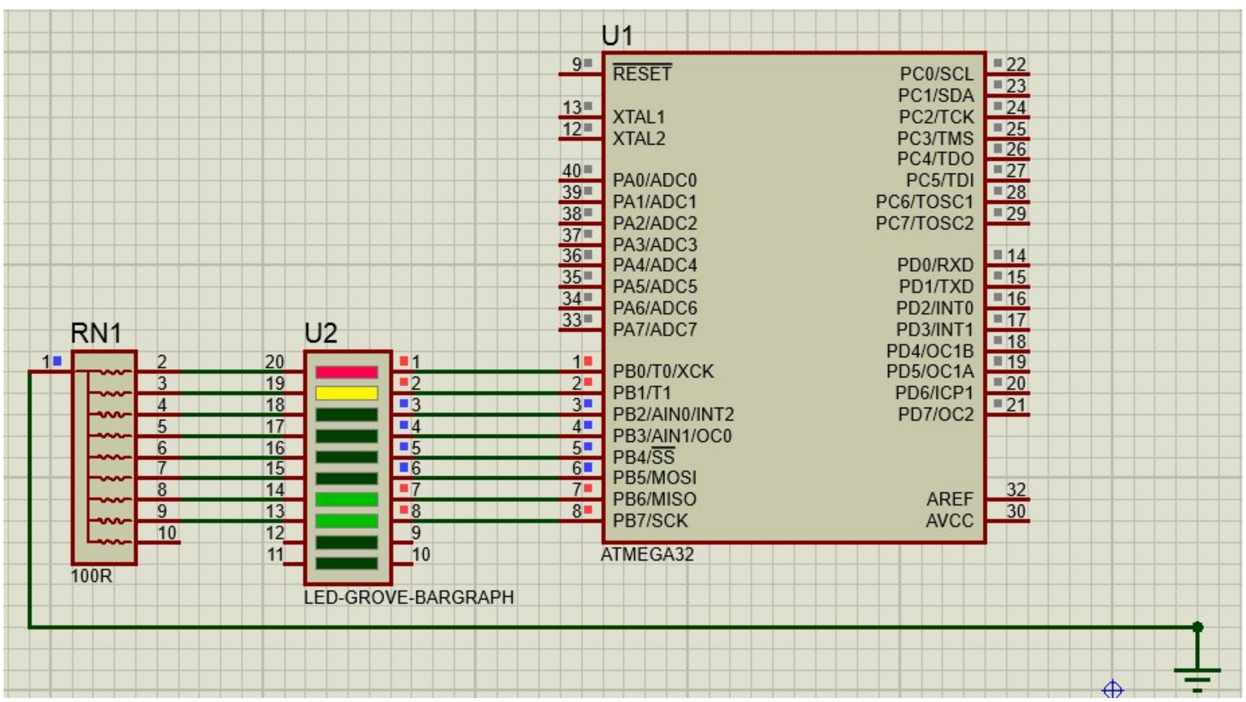


Figure 1: digit = '2'

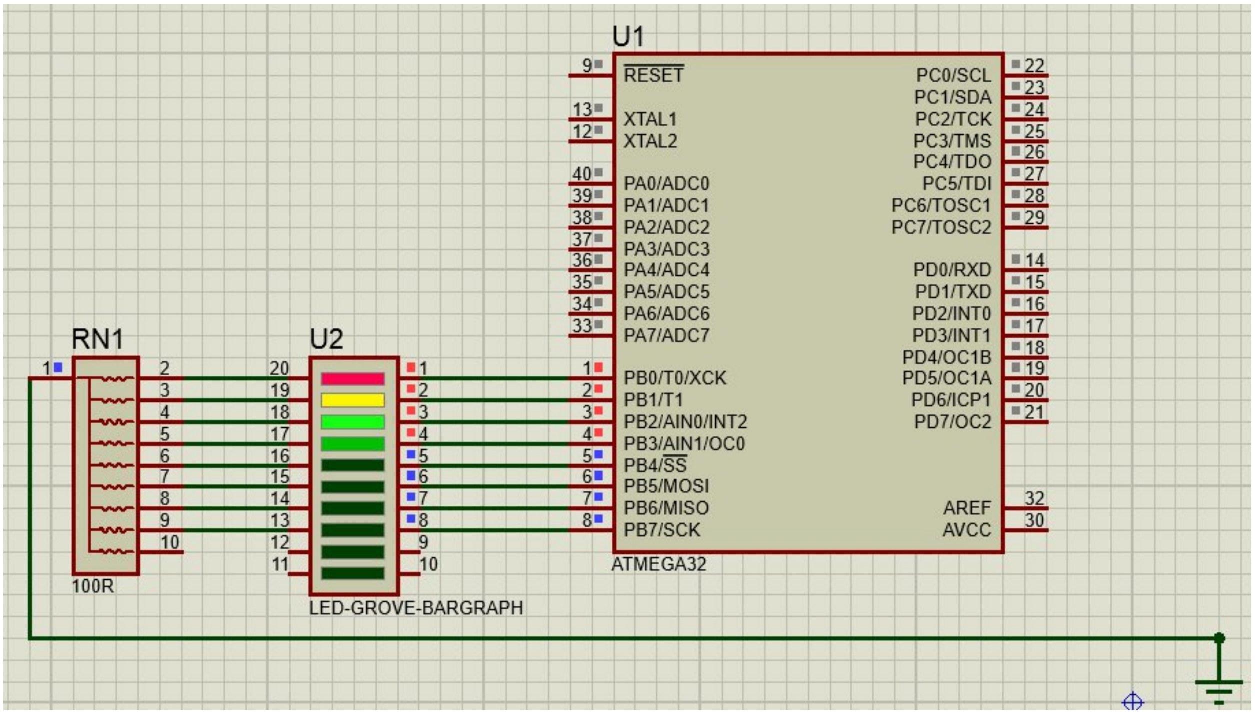


Figure 2: digit = '1'

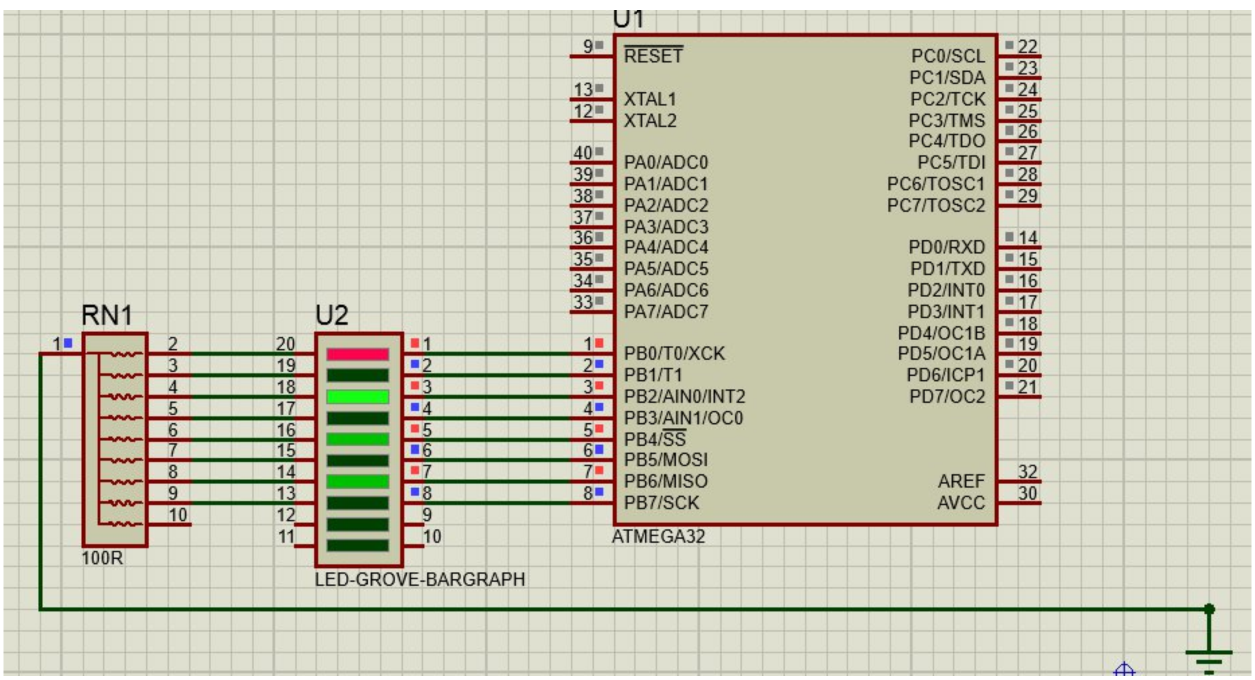


Figure 3: digit = '3'

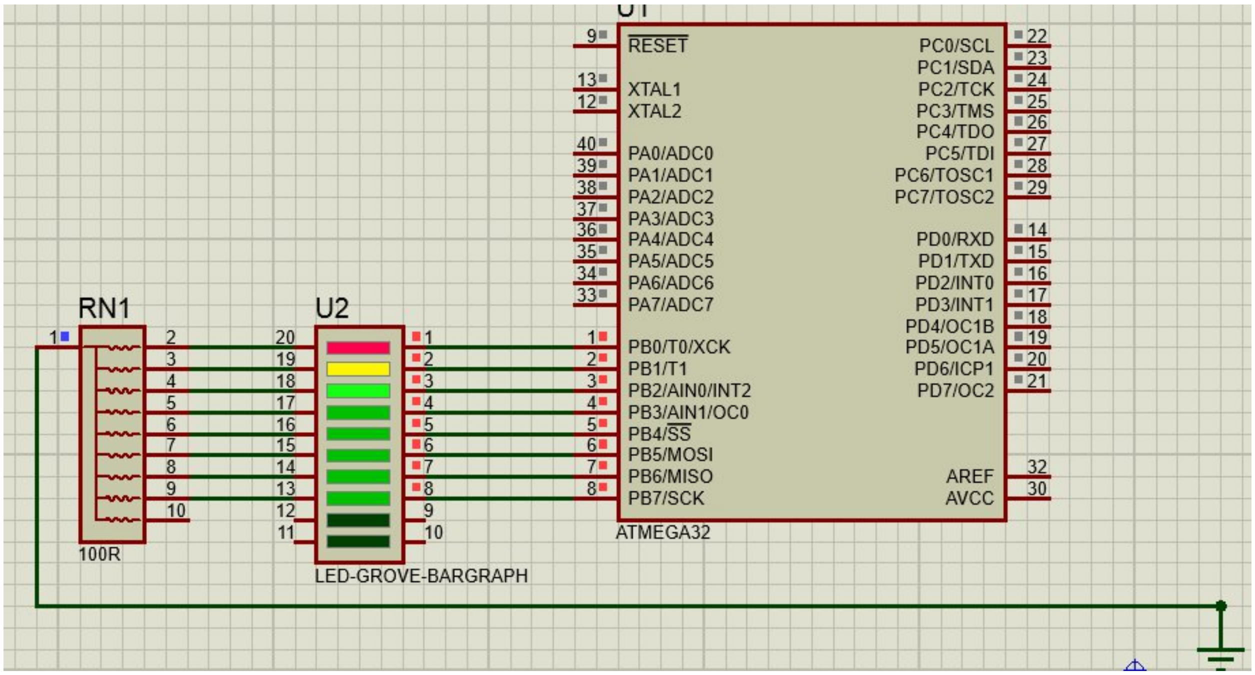
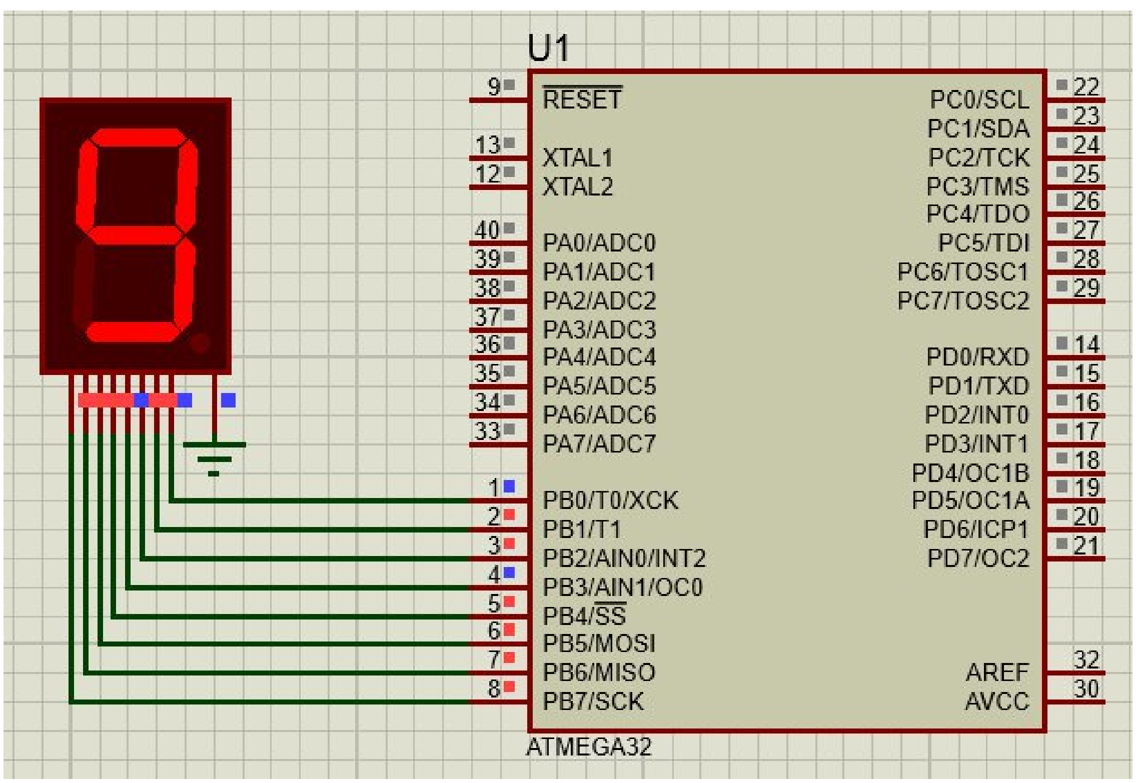


Figure 4: digit =''*

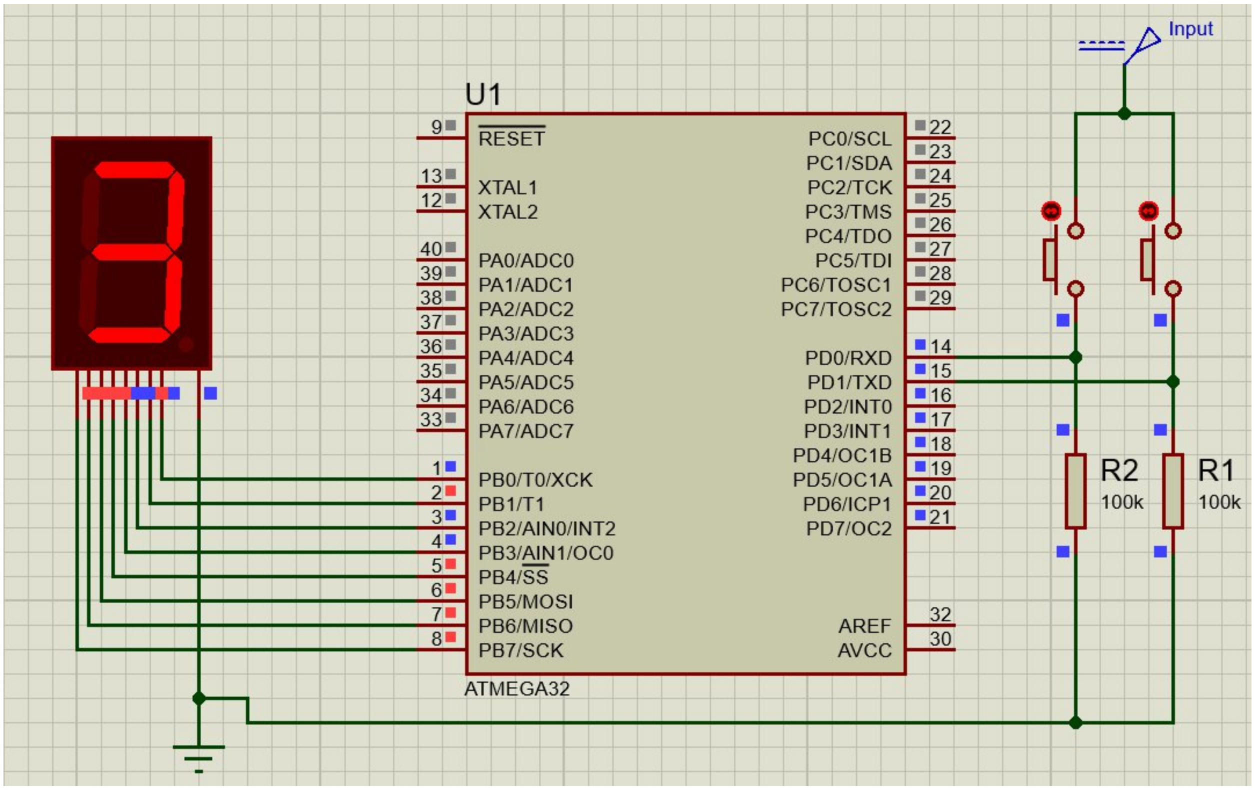
Task 4

```
#define F_CPU 8000000UL // 8 MHz clock frequency
#include <avr/io.h>
#include <util/delay.h>

int main(void)
{
    DDRB = 0xFF;    // set PORTB for Output
    int a = 0;
    do
    {
        switch (a){
            case 0: PORTB = 0b11111100; break;
            case 1: PORTB = 0b01100000; break;
            case 2: PORTB = 0b11011010; break;
            case 3: PORTB = 0b11110010; break;
            case 4: PORTB = 0b01100110; break;
            case 5: PORTB = 0b10110110; break;
            case 6: PORTB = 0b10111110; break;
            case 7: PORTB = 0b11100000; break;
            case 8: PORTB = 0b11111110; break;
            case 9: PORTB = 0b11110110; break;
        }
        a = a+1;
        _delay_ms(1000);
    } while (a<10);
    return 1;
}
```



For increment and decrement through 2 push buttons

```c
#define F_CPU 8000000UL // 8 MHz clock frequency
#include <avr/io.h>
#include <util/delay.h>

int main(void)
{
    DDRB = 0xFF;    // set PORTB for Output
    DDRD = 0b11111100; // set 0 and 1 as input inPORTD
    int a = 0;
    do
    {
        switch (a){
            case 0: PORTB = 0b11111100; break;
            case 1: PORTB = 0b01100000; break;
            case 2: PORTB = 0b11011010; break;
            case 3: PORTB = 0b11110010; break;
            case 4: PORTB = 0b01100110; break;
            case 5: PORTB = 0b10110110; break;
            case 6: PORTB = 0b10111110; break;
            case 7: PORTB = 0b11100000; break;
            case 8: PORTB = 0b11111110; break;
            case 9: PORTB = 0b11110110; break;
        }
        if (PIND==1){
            _delay_ms(500);
            if (a!=9){a = a+1;}
            else
                a=0;
        }
        if (PIND==2)
        {   _delay_ms(500);
            if (a!=0){a = a-1;}
            else
                a=9;
        }
    } while (a<10);
    return 1;
}
```


Input
U1
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
ATMEGA32
R2
100k
R1
100k